

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	
Team ID	
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below along with the component and characteristic details in Table-1 and Table-2.

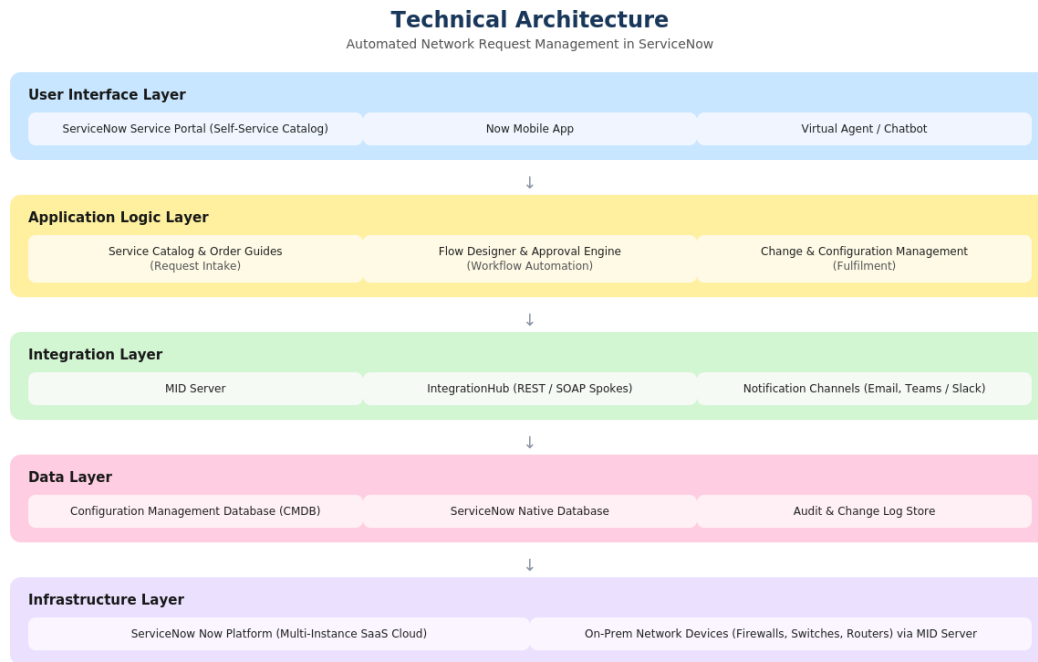


Figure 1: Technical architecture — Automated Network Request Management in ServiceNow

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the requester and approvers interact with the system	ServiceNow Service Portal, Now Mobile App, Virtual Agent
2.	Application Logic-1	Request intake and catalog management	ServiceNow Service Catalog, Order Guides, Glide Script
3.	Application Logic-2	Approval workflow and routing automation	ServiceNow Flow Designer, Approval Engine
4.	Application Logic-3	Change and configuration management	ServiceNow Change Management, CMDB (CSDM)
5.	Database	Platform data storage and configuration	ServiceNow native database
6.	Cloud Database	Database service on cloud	ServiceNow multi-instance cloud database
7.	File Storage	Storage for attachments (diagrams, approvals)	ServiceNow Attachment API / Now Platform storage
8.	External API-1	Provisioning on network devices	REST APIs to firewall / switch / router management systems via MID Server
9.	External API-2	Notifications to requesters and approvers	Email (SMTP), Microsoft Teams / Slack via IntegrationHub spokes
10.	Machine Learning Model	Risk scoring for change requests	ServiceNow Predictive Intelligence

11.	Infrastructure (Server / Cloud)	Application deployment	ServiceNow Now Platform SaaS instance; on-prem MID Server for network integration
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Table-2: Application Characteristics

Characteristics	Description	Technology
Open-Source Frameworks	Front-end frameworks used within the Service Portal	AngularJS, Bootstrap
Security Implementations	Access controls and data protection measures	Role-based ACLs, OAuth 2.0, TLS 1.2+, encryption at rest
Scalable Architecture	Multi-tenant SaaS architecture with modular workflows	ServiceNow Now Platform (multi-instance SaaS)
Availability	Redundant, load-balanced platform architecture	ServiceNow high-availability architecture, 99.9% SLA
Performance	Caching and asynchronous execution for high request volumes	IntegrationHub caching, Flow Designer asynchronous flows